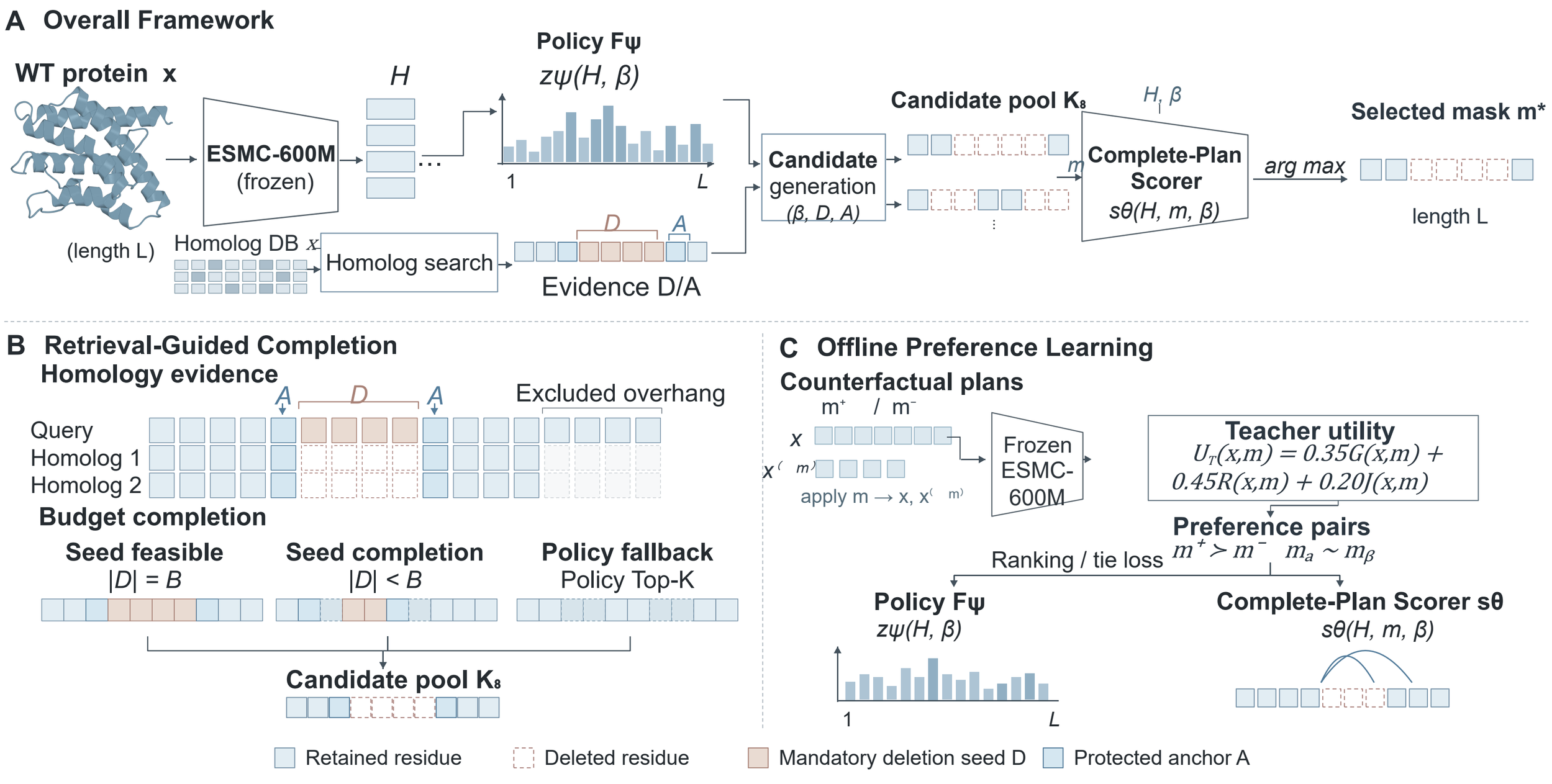

A Overall Framework
WT protein x
(length L)
ESMC-600M
(frozen)
H
Policy Fψ
zψ(H, β)
1
L
Homolog DB x
Homolog search
D
A
Evidence D/A
Candidate
generation
(β, D, A)
Candidate pool K8
m
H, β
Complete-Plan
Scorer
sθ(H, m, β)
arg max
Selected mask m*
length L
B Retrieval-Guided Completion
Homology evidence
A
D
A
Excluded overhang
Query
Homolog 1
Homolog 2
Budget completion
Seed feasible
|D| = B
Seed completion
|D| < B
Policy fallback
Policy Top-K
Candidate pool K8
C Offline Preference Learning
Counterfactual plans
m+ / m−
x
x(m)
apply m → x, x(m)
Frozen
ESMC-
600M
Teacher utility
UT(x,m) = 0.35G(x,m) +
0.45R(x,m) + 0.20J(x,m)
Preference pairs
m+ ≻ m− ma ~ mβ
Ranking / tie loss
Policy Fψ
zψ(H, β)
1
L
Complete-Plan Scorer sθ
sθ(H, m, β)
Retained residue
Deleted residue
Mandatory deletion seed D
Protected anchor A